

Robust Stability and performance analysis using LQR and H_∞ for a Two-Wheeled Self-Balancing Robot

Bimlendra Ray

Department of Mechanical Engineering

The University of Texas at Dallas

Richardson, TX, USA

Email: bxr240009@utdallas.edu

Abstract—The two wheeled self-balancing robot (TWSBR) is stabilized using linear quadratic regulator (LQR) and mixed-sensitivity H_∞ controller. Along with that, the robustness and performance of each controller is studied. The controllers are evaluated using nominal initial-condition response, sinusoidal disturbance rejection, structured mass and height uncertainty, worst-case gain analysis, and Monte Carlo simulation over uncertain physical parameters. The LQR controller provides a simple full-state feedback baseline, while the H_∞ controller explicitly shapes sensitivity, control effort, and high-frequency complementary sensitivity. The simulations show how the two methods trade nominal response, voltage usage, disturbance attenuation, and robustness to model uncertainty.

Index Terms—self-balancing robot, LQR, H_∞ control, robust control, mixed sensitivity, Monte Carlo simulation

I. INTRODUCTION

TWSBR is a particular type of robot, usually consisting of two wheels and some sensors that can adjust to maintain a balanced state []. Two-wheeled self-balancing robots are nonlinear, underactuated systems whose upright equilibrium is open-loop unstable. Around the upright operating point, the balancing dynamics can be represented by a linear state-space model and stabilized using feedback control. Classical optimal control, such as LQR, is attractive because it gives a direct state-feedback law with transparent state and input penalties. However, performance can degrade under plant-parameter variations. Robust control methods such as H_∞ synthesis can improve worst-case behavior by explicitly shaping closed-loop sensitivity and control effort.

This work studies a common-mode balancing input for a two-wheeled self-balancing robot. This report consist of the nominal comparison, disturbance-rejection experiment, parametric-uncertainty experiment, and Monte Carlo simulation. The main objective is to compare LQR and H_∞ controllers under increasingly uncertain operating conditions like parametric uncertainty.

The main contributions are:

- A one-input common-mode balancing model is used so that LQR and H_∞ controllers are compared.
- Two parametric uncertainty sources are considered: payload-mass variation and body-height variation.

- The comparison includes nominal stabilization, disturbance rejection, robust performance, actuator voltage usage, and Monte Carlo validation.
- The tradeoff between tight pitch regulation and actuator-safe robust behavior is explicitly evaluated.

II. ROBOT MODEL

The balancing subsystem is represented using the state vector

$$x = [\theta \quad \psi \quad \dot{\theta} \quad \dot{\psi}]^T, \quad (1)$$

where θ is the average wheel angle and ψ is the body pitch angle. The nominal linear model used in the first two experiments is

$$\dot{x} = Ax + Bu, \quad (2)$$

with

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 55.540 & -0.610 & 0.610 \\ 0 & 62.794 & -0.316 & 0.316 \end{bmatrix}, \quad (3)$$

and the original two-motor input matrix

$$B_2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 9.385 & 9.385 \\ -4.857 & -4.857 \end{bmatrix}. \quad (4)$$

For balancing, the two motor voltages are commanded equally:

$$u_b = v_l + v_r, \quad v_l = v_r = \frac{u_b}{2}. \quad (5)$$

Therefore, the effective one-input model uses

$$B = B_2(:, 1) = [0 \quad 0 \quad 9.385 \quad -4.857]^T. \quad (6)$$

The output matrix is selected as $C = I_4$, so full-state feedback and full-state performance outputs are available.

TABLE I
NOMINAL PHYSICAL PARAMETERS USED IN THE UNCERTAINTY MODEL

Parameter	Symbol	Nominal Value
Payload/body mass	M_0	0.25 kg
Robot height	H_0	0.17 m
Wheel radius	R	0.0325 m
Wheelbase width	W	0.192 m
Body depth	D	0.082 m
Gear ratio	n	30
Motor resistance	R_m	2.9 Ω
Back-emf constant	K_b	0.024
Torque constant	K_t	0.025

III. UNCERTAINTY DESCRIPTION AND ROBUST OBJECTIVE

The presentation identifies two major uncertainty sources and one sinusoidal disturbance for the balancing problem. First, payload changes alter the total mass and shift the inertial properties or center of mass (COM). Second, height changes modify the center-of-mass location and therefore the gravitational torque about the wheel axis. Third, external disturbances act through the acceleration channels and represent pushes, floor irregularities, or unmodeled input effects. The uncertainties in the plant is shown in the Figure 1

The robust-control objective is therefore not only to stabilize the nominal plant. The closed-loop system should remain stable for the full modeled parameter range, keep pitch and wheel states bounded during disturbances, and satisfy practical actuator limits. In this project the motor-voltage limit is treated as

$$|v_l|, |v_r| \leq 24 \text{ V}. \quad (7)$$

This actuator-safety requirement is important because a controller that gives excellent pitch regulation but demands unrealistic voltage is not physically implementable.

IV. CONTROLLER DESIGN

A. LQR Controller

The LQR controller minimizes

$$J = \int_0^\infty (x^T Q x + u^T R u) dt, \quad (8)$$

and applies the state-feedback law

$$u_b = -K_{\text{LQR}} x. \quad (9)$$

The nominal simulations use

$$Q = \text{diag}(10000, 1, 10000, 1), \quad (10)$$

with $R = 200$ for the nominal initial-condition experiment and $R = 500$ for the disturbance and uncertainty experiments. The use of the different input penalty is because due to uncertainty it requires more input voltage to stabilize the system but we have constraints and we want to reduce the effort as much as possible. The large penalties on θ and $\dot{\theta}$ are used to regulate wheel motion while stabilizing body pitch.

LQR is useful as a nominal baseline, but it does not directly provide uncertainty guarantees, frequency-dependent performance shaping, or worst-case performance bounds. Actuator limits can only be handled indirectly by increasing R , which reduces control effort but may also reduce regulation performance.

B. Mixed-Sensitivity H_∞ Controller

The H_∞ controller is synthesized using MATLAB's `mixsyn` command. The design solves the mixed-sensitivity problem

$$\min_K \left\| \begin{bmatrix} W_s S \\ W_u K S \\ W_t T \end{bmatrix} \right\|_\infty, \quad (11)$$

where

$$S = (I + GK)^{-1}, \quad T = GK(I + GK)^{-1}. \quad (12)$$

The sensitivity weight has the form

$$W_{s,\text{base}}(s) = \frac{s/M_s + \omega_b}{s + \omega_b A_s}, \quad (13)$$

with $M_s = 3$, $A_s = 0.15$, and $\omega_b = 4$ rad/s. The weighting matrix penalizes pitch-related channels more strongly:

$$W_s = \text{blkdiag}(0.01W_{s,\text{base}}, 0.1W_{s,\text{base}}, 0.005W_{s,\text{base}}, 0.08W_{s,\text{base}}). \quad (14)$$

The control weight is selected from the motor-voltage limit:

$$W_u = \frac{1}{4u_{\max}}, \quad u_{\max} = 24 \text{ V}, \quad (15)$$

with a 4x scaling in the uncertainty study. The complementary-sensitivity weight is

$$W_t = 0.01 \frac{s + \omega_t/M_t}{A_t s + \omega_t} I_4, \quad (16)$$

where $M_t = 3$, $A_t = 5 \times 10^{-2}$, and $\omega_t = 50$ rad/s.

Whereas the $W_s S$ channel shapes disturbance rejection and low-frequency tracking behavior. The $W_u K S$ channel penalizes control effort and helps keep the motor voltage practical. The $W_t T$ channel limits high-frequency amplification and improves noise robustness. The achieved H_∞ value γ is used to evaluate how closely the weighted design specifications are satisfied. Using this weights, the obtained γ value is 0.667. The final bode plot shown in the Figure 2 satisfy the requirements of $\gamma < 1$ to achieve robust performance for uncertainty in M and H.

V. SIMULATION EXPERIMENTS

A. Experiment 1: Nominal Initial-Condition Response

The first experiment compares the nominal closed-loop response of LQR and H_∞ control over 10 s with

$$x(0) = [0 \quad 0.1 \quad 0.3 \quad 0.3]^T. \quad (17)$$

The response variables are average wheel angle, wheel angular velocity, body pitch angle, body pitch angular velocity, and left/right motor voltages. From the Fig. 3, the wheel angle and

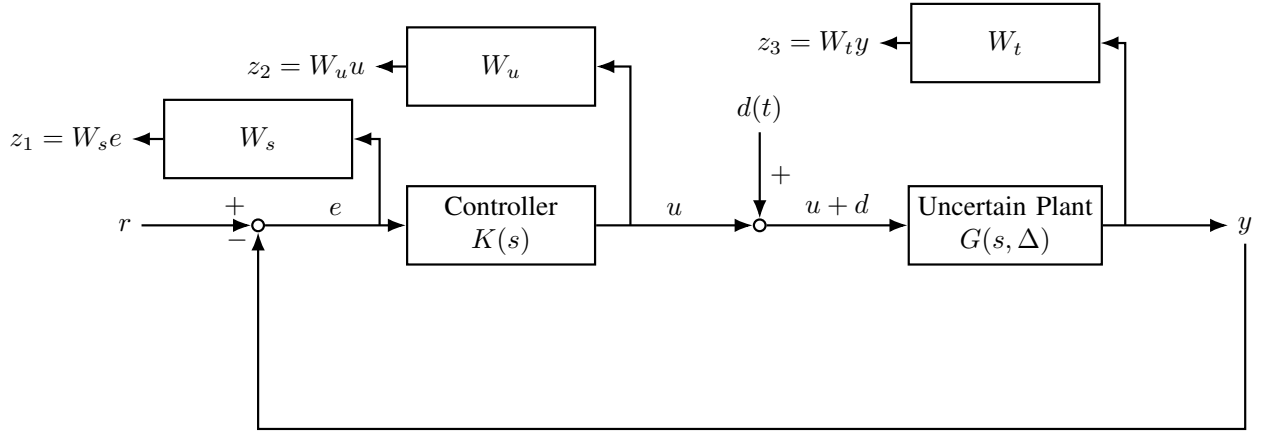


Fig. 1. Robust-control architecture for the mixed-sensitivity H_∞ design.

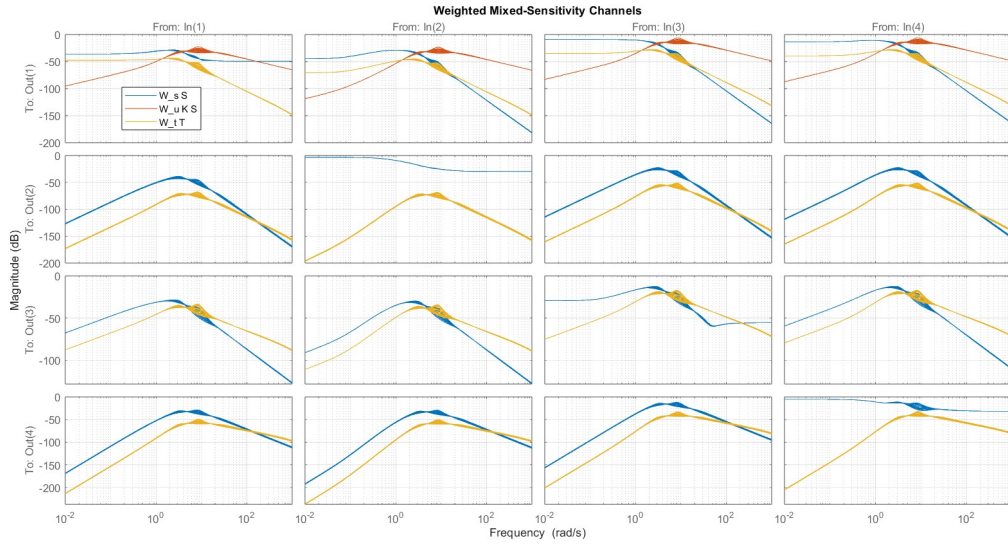


Fig. 2. Mixed-sensitivity frequency-response for $W_s S$, $W_u K S$, and $W_t T$ for uncertainty in M and H.

pitch angle fluctuation is smaller for LQR controller compared to the H_∞ controller. However, pitch angle velocity is higher for LQR compared to H_∞ . The input required for LQR is comparatively very high, 16V, where as for H_∞ controller it is less, 3V. This shows that LQR is better at stabilizing the system at the expense of input but H_∞ is better at optimizing the use of input and smoothly stabilizing the system with some fluctuation.

B. Experiment 2: Sinusoidal Disturbance Rejection

During the self-balancing control process of TWSBR, external impacts and other interferences make it easy to be disturbed, causing the load to shake unexpectedly. The purpose of this experiment is to verify the robustness of LQR and H_∞ controller. The experiment injects an external disturbance into the acceleration states:

$$\dot{x} = Ax + Bu + B_d f(t), \quad (18)$$

where

$$B_d = \begin{bmatrix} 0 & 0 & 1 & 1 \end{bmatrix}^T, \quad f(t) = 0.3 \sin(t). \quad (19)$$

The same initial condition as Experiment 1 is used. The LQR closed-loop system is simulated as

$$\dot{x} = (A - BK_{LQR})x + B_d f(t), \quad (20)$$

while the H_∞ controller is simulated using the augmented dynamic closed-loop model. Figure 4 shows that both of the controller able to stabilize the system with fluctuation. The LQR is still able to stabilize the system very well and well damped but require more input volt compared to the H_∞ .

C. Experiment 3: Parametric Uncertainty

The third experiment evaluates robustness to physical parameter variation. The controllers are designed at the nominal values

$$M_0 = 0.25 \text{ kg}, \quad H_0 = 0.17 \text{ m}, \quad (21)$$

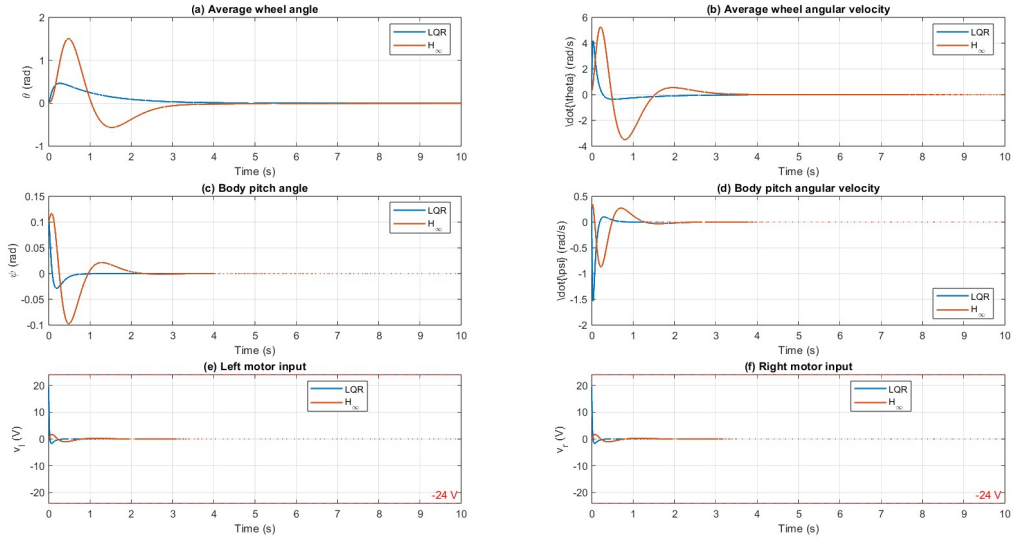


Fig. 3. Nominal response of the plant using LQR and H_∞ controller.

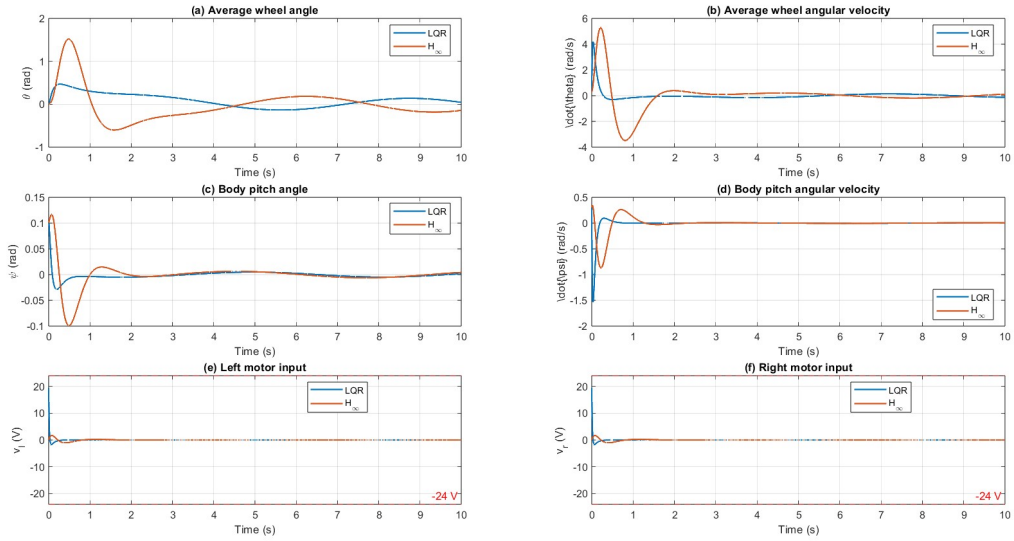


Fig. 4. Disturbance-rejection response comparing LQR and H_∞ controllers for additive disturbance in the input.

and then held fixed while the plant changes over

$$M \in [0.2, 0.3] \text{ kg}, \quad H \in [0.14, 0.20] \text{ m}. \quad (22)$$

The uncertain balancing model is generated from the physical parameters M and H by first defining the center-of-mass length

$$L = \frac{H}{2}. \quad (23)$$

The wheel and body inertias are modeled as

$$J_w = \frac{mR^2}{2}, \quad J_\psi = \frac{ML^2}{3}, \quad J_\phi = \frac{M(W^2 + D^2)}{12}. \quad (24)$$

The motor constants are grouped as

$$\alpha = \frac{nK_t}{R_m}, \quad \beta = \frac{nK_tK_b}{R_m} + f_m. \quad (25)$$

The coupled inertia terms are

$$E_{11} = (2m + M)R^2 + 2J_w + 2n^2J_m, \quad (26)$$

$$E_{12} = MLR - 2n^2J_m, \quad (27)$$

$$E_{22} = ML^2 + J_\psi + 2n^2J_m, \quad (28)$$

with

$$\Delta_E = E_{11}E_{22} - E_{12}^2. \quad (29)$$

Using these quantities, the balancing subsystem is written as

$$\dot{x} = A(M, H)x + B(M, H)u_b, \quad (30)$$

where

$$A(M, H) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & A_{32} & A_{33} & A_{34} \\ 0 & A_{42} & A_{43} & A_{44} \end{bmatrix}, \quad (31)$$

and

$$A_{32} = -\frac{gMLE_{12}}{\Delta_E}, \quad (32)$$

$$A_{42} = \frac{gMLE_{11}}{\Delta_E}, \quad (33)$$

$$A_{33} = -\frac{(2\beta + f_w)E_{22} + 2\beta E_{12}}{\Delta_E}, \quad (34)$$

$$A_{43} = \frac{(2\beta + f_w)E_{12} + 2\beta E_{11}}{\Delta_E}, \quad (35)$$

$$A_{34} = \frac{2\beta(E_{22} + E_{12})}{\Delta_E}, \quad (36)$$

$$A_{44} = -\frac{2\beta(E_{11} + E_{12})}{\Delta_E}. \quad (37)$$

The two motor inputs enter symmetrically through

$$B_2(M, H) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ B_3 & B_3 \\ B_4 & B_4 \end{bmatrix}, \quad (38)$$

where

$$B_3 = \frac{\alpha(E_{22} + E_{12})}{\Delta_E}, \quad (39)$$

$$B_4 = -\frac{\alpha(E_{11} + E_{12})}{\Delta_E}. \quad (40)$$

For the common-mode balancing input, the effective one-input matrix is

$$B(M, H) = B_2(:, 1) = \begin{bmatrix} 0 \\ 0 \\ B_3 \\ B_4 \end{bmatrix}. \quad (41)$$

Thus, payload mass and robot height enter the state-space model through L , J_ψ , J_ϕ , E_{11} , E_{12} , and E_{22} , which directly modify the gravitational, damping, and input-coupling terms of the balancing dynamics. The uncertain plant is modeled with `ureal` objects and analyzed using worst-case gain computations. The robust stability analysis using `robstab` showed that both controllers remained robustly stable for the entire modeled uncertainty set. The LQR controller achieved a robust stability margin of approximately, $\mu_{LQR} = 3.93$ while the H_∞ controller achieved $\mu_{H_\infty} = 1.66$. Since both margins are greater than one, no admissible uncertainty combination within the modeled range destabilizes the closed-loop system.

The destabilizing substitutions returned by `robstab` occurred outside the original uncertainty ranges. For the LQR controller, instability was predicted approximately near

$$M \approx 0.447 \text{ kg}, \quad H \approx 0.288 \text{ m}, \quad (42)$$

whereas for the H_∞ controller the destabilizing point was approximately

$$M \approx 0.333 \text{ kg}, \quad H \approx 0.220 \text{ m}. \quad (43)$$

This result shows that the LQR controller can withstand a larger level of modeled uncertainty before the closed-loop system approaches instability. The sustained oscillatory behavior observed in the worst-case simulations in Fig. 5 is caused by the dominant complex pole pairs moving toward the imaginary axis, as summarized in Table IV. This pole migration highlights the effect of parametric uncertainty on the balancing dynamics of the TWSBR. Specifically, increasing the payload mass M and center-of-mass height H increases the gravitational destabilizing torque, which reduces damping in the dominant closed-loop modes and moves the system closer to the stability boundary.

Time-domain simulations were performed under nominal, mass-varying, height-varying, and combined uncertainty conditions. The LQR controller produced tighter pitch-angle regulation, with the maximum pitch deviation remaining close to

$$|\psi|_{\max} \approx 0.10 \text{ rad} \quad (44)$$

for all uncertainty cases within the given range. In comparison, the H_∞ controller exhibited larger pitch deviations as uncertainty increased:

$$0.115 \text{ rad} \leq |\psi|_{\max} \leq 0.179 \text{ rad}. \quad (45)$$

However, the LQR controller achieved this tighter regulation using significantly more aggressive control action. The maximum each wheel voltage for LQR remained approximately

$$|v|_{\max} \approx 18.9 \text{ V}, \quad (46)$$

which is close to the actuator limit of 24 V. In contrast, the H_∞ controller required substantially smaller actuator effort, with the maximum each wheel voltage remaining below

$$|v|_{\max} \approx 2.64 \text{ V}, \quad (47)$$

even under the combined worst modeled uncertainty.

Furthermore, to evaluate the robustness of the self-balancing robot controllers, four endpoint simulation scenarios were considered by varying the payload mass M and robot height H . These cases represent the nominal and extreme combinations of the modeled uncertainty range. For each case, the LQR and H_∞ controllers were kept fixed and the closed-loop response was simulated from the same initial condition.

TABLE II
SIMULATION SCENARIOS FOR PARAMETRIC UNCERTAINTY STUDY

Scenario	Mass M (kg)	Height H (m)
Nominal	0.25	0.17
Mass change	0.30	0.17
Height change	0.25	0.20
Both changes	0.30	0.20

These parameter combinations were selected to test the closed-loop behavior at the endpoints of the uncertainty set.

The “Mass change” case isolates the effect of increasing the robot mass, the “Height change” case isolates the effect of increasing the center-of-mass height, and the “Both changes” case represents the simultaneous worst endpoint of both uncertain parameters.

For both controllers, the pitch-angle response exhibits sustained oscillatory behavior near the boundary of instability as shown in Figure 5. These oscillations occur because the uncertainty scaling has been increased close to the robust-stability margin. The LQR worst-case substitution produces higher-frequency oscillation, while the H_∞ controller oscillates at a lower frequency corresponding to its smaller robust stability margin. The associated motor-voltage responses remain bounded but show persistent periodic control action required to maintain marginal stability. Importantly, these destabilizing substitutions occur outside the original modeled uncertainty set because both controllers have robust stability margins greater than one. Therefore, the simulations represent near-instability conditions rather than actual operating conditions.

The results show that both controllers remain within actuator limits for all modeled uncertainty cases as shown in the Figure 6. However, the LQR controller consistently requires larger transient control effort compared to the H_∞ controller. The mixed-sensitivity H_∞ controller produces significantly smaller voltage demand because the control-effort weighting W_u explicitly penalizes aggressive actuator usage during synthesis.

The LQR controller produces tighter pitch regulation with faster damping and smaller overshoot across all uncertainty realizations as shown in the Figure 7. This behavior is expected because LQR uses high penalty for pitch angle. The H_∞ controller exhibits larger transient oscillations and slower settling, especially when both mass and height are simultaneously increased. This occurs because the robust controller sacrifices aggressive regulation to maintain robustness and reduced actuator demand. As the center-of-mass height increases, the pendulum dynamics becomes more unstable due to increased gravitational torque. This effect is visible in the larger oscillatory response of the H_∞ controller during the height-change and combined-uncertainty cases. Despite the increased oscillation, all responses remain bounded and converge to equilibrium, confirming closed-loop stability throughout the modeled uncertainty set.

The wheel-angle response illustrates the balancing motion required to stabilize the robot body in Figure 8. The H_∞ controller allows significantly larger wheel displacement compared to LQR, especially during transient balancing maneuvers. This behavior reflects a key tradeoff in robust control design. Instead of generating large motor voltages to quickly suppress pitch motion, the H_∞ controller permits larger wheel motion while maintaining low actuator effort. The LQR controller produces smaller wheel displacement due to its aggressive feedback action. However, this tighter regulation is achieved through larger control gains and increased voltage demand because of higher penalty given for pitch angle. The simulations show that increased payload mass and center-

TABLE III
EXPERIMENT 3 PARAMETRIC-UNCERTAINTY METRICS

Metric	LQR	H_∞
Max $ \psi $ (rad)	0.1005	0.1789
Max $ \theta $ (rad)	0.9458	2.8196
Max $ u_b $ (V)	28.1490	5.2840
Max $ v $ (V)	14.0745	2.6420
Robust stability lower bound	4.1061	1.6609
Robust stability upper bound	4.1169	1.6642
Worst-case gain lower bound	1.3703	1.1158×10^4
Worst-case gain upper bound	1.4240	1.8362×10^5

TABLE IV
WORST-CASE DOMINANT POLE ANALYSIS

Controller	Dominant poles	Frequency (rad/s)	Interpretation
LQR	$\lambda = \pm j18.8233$	18.8233	Marginal high-frequency mode
H_∞	$\lambda = \pm j8.6242$	8.6242	Marginal lower-frequency mode

of-mass height require larger wheel motion for stabilization, particularly for the robust controller.

D. Monte Carlo Uncertainty Simulation

The Monte Carlo simulations samples uncertain mass and height values and simulates both controllers for each sample. For each run, the following metrics are recorded:

$$\max_t |\psi(t)|, \quad \max_t |v(t)|, \quad (48)$$

along with closed-loop stability. The script uses $N_{MC} = 10000$ samples and overlays nominal pitch response on the sampled trajectories (Fig. 9).

VI. CONCLUSION

This open loop unstable system is controlled using feedback controller like LQR and H_∞ . For the nominal dynamics, it is stabilized very well using LQR method and it also shows a good performance for external disturbance. For uncertain parameter, LQR is able stabilize the system well but control input required is high whereas H_∞ controller applied very small amount of control input and able to stabilize the system with some overshoot. So, in terms of minimum control input, H_∞ shows better performance while maintaining the robust stability and robust performance condition.

REFERENCES

- [1] M. Mahmoudian, A. Ahmadi, and M. B. Menhaj, “Robust control of a two-wheeled self-balancing robot using mixed sensitivity H_∞ and loop shaping design,” in Proc. 6th RSI Int. Conf. Robotics and Mechatronics (ICRoM), Tehran, Iran, 2018, pp. 316–321.

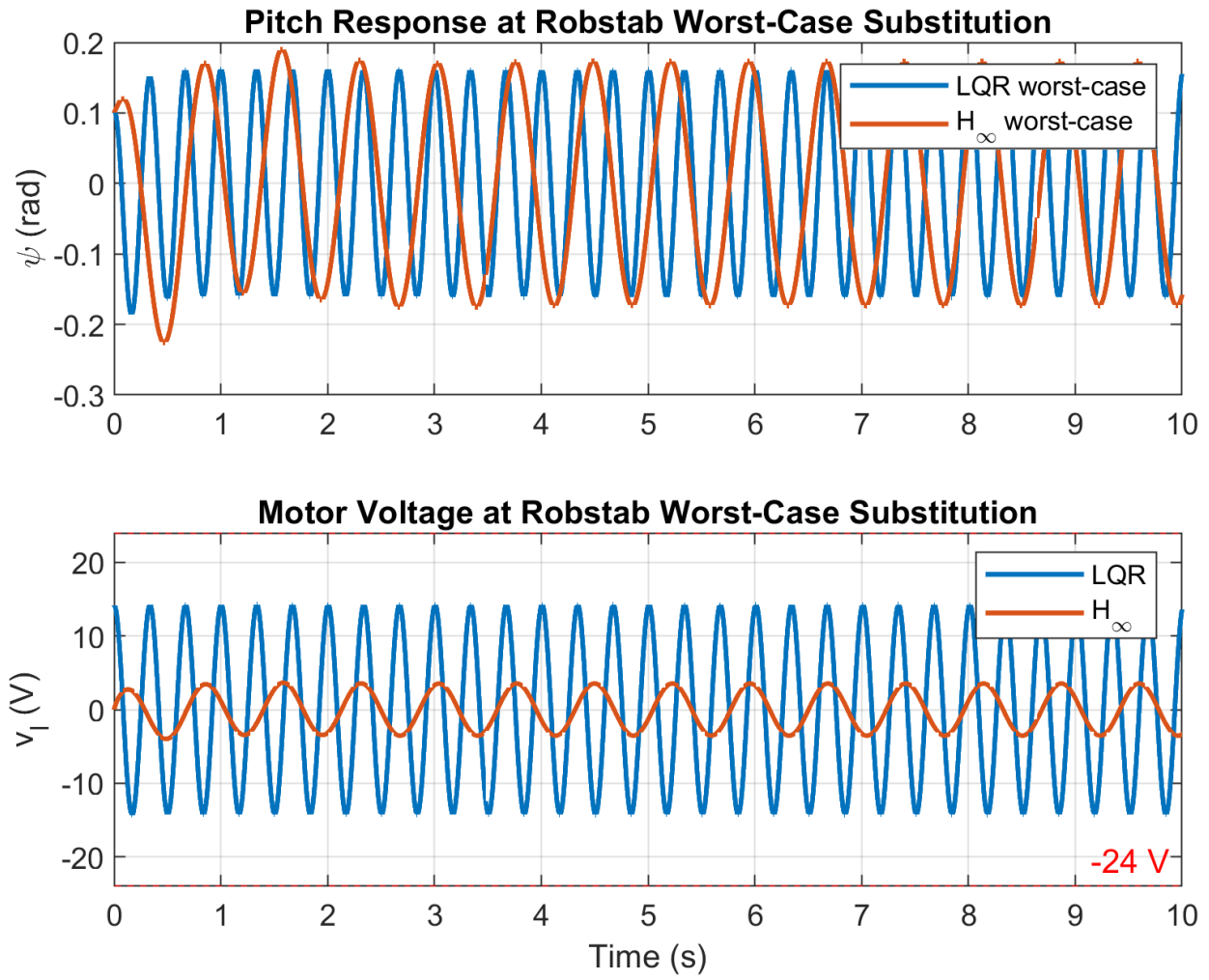


Fig. 5. Simulation done at destabilizing point for LQR and H_∞ for pitch rate and control input.

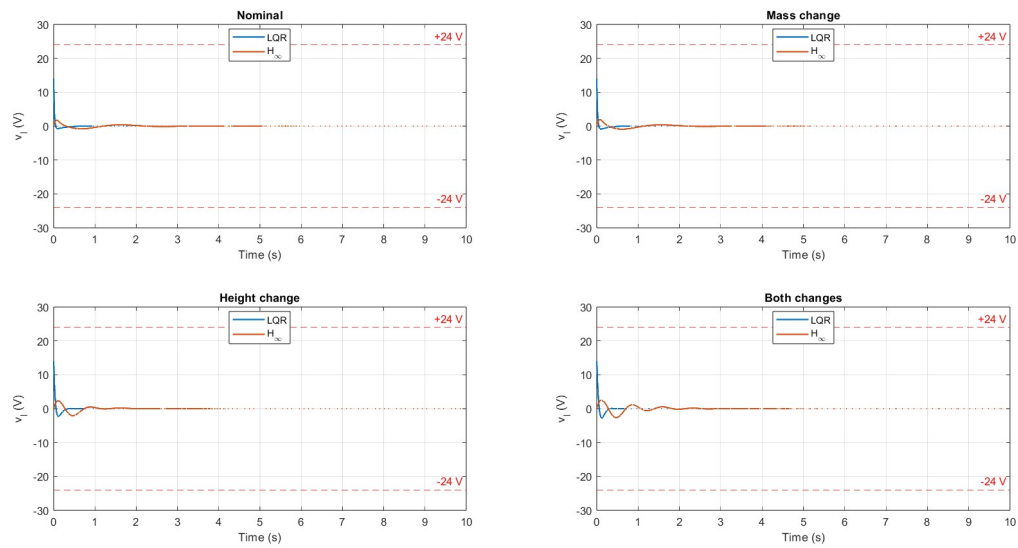


Fig. 6. Motor-voltage responses under uncertain parameter variations.

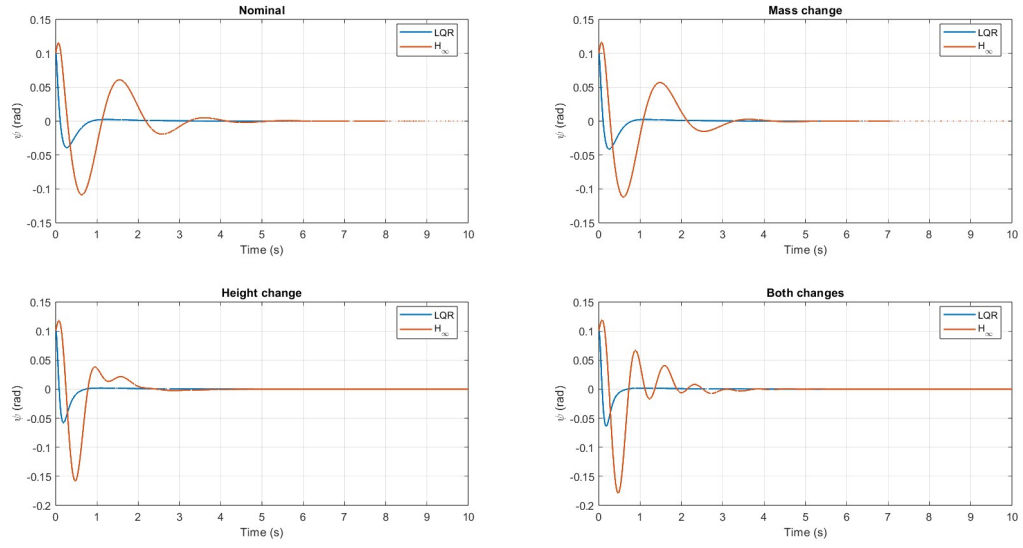


Fig. 7. Pitch-angle responses under uncertain parameter variations.

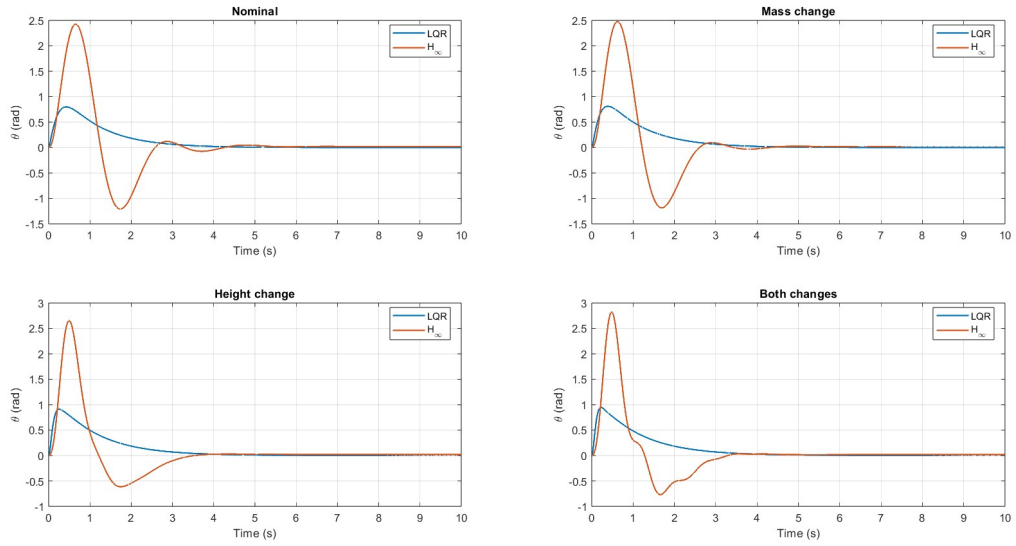


Fig. 8. Wheel-angle responses under uncertain parameter variations.

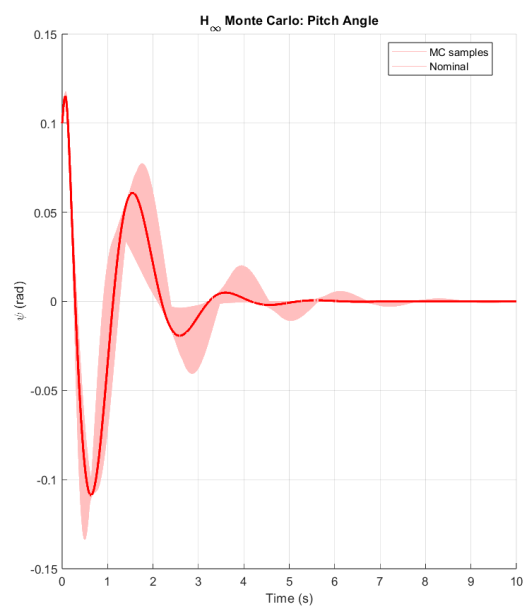
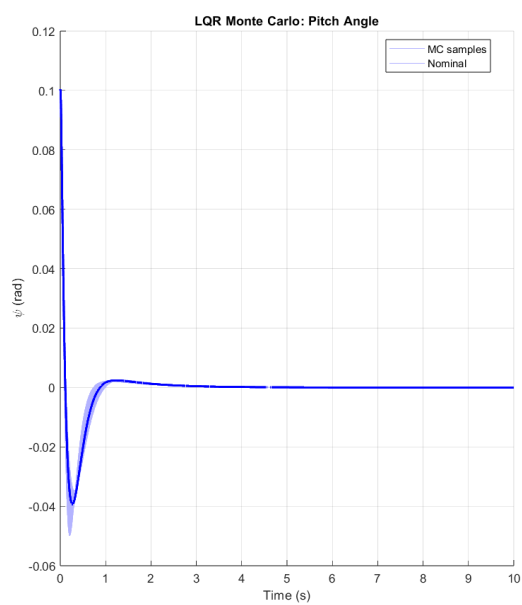


Fig. 9. Monte-carlo simulation. $N=10000$